# An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

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CyberAgent, Inc.  Chiba University

STOC 2027 submission  (author video for the PC)

# Outline

# Model checking: $\mathrm{FO}^3$ is the exception

| logic | model checking |
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| $\mathrm{FO}^k$, $k \geq 4$ | no $O(n^{k-\varepsilon})$ (hyperclique hypothesis) |
| $\mathrm{FO}^3$ | $2^{O(\lvert\varphi\rvert)} \cdot n^{\omega}$ (**nakamura-uezato2027**) |

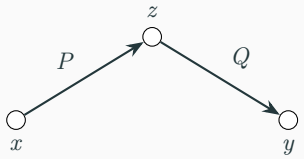



$\mathrm{FO}^3$ and CoR define the same binary queries (**tarski-givant1987**)

The factor $2^{O(|\varphi|)}$ comes from the translation into CoR.

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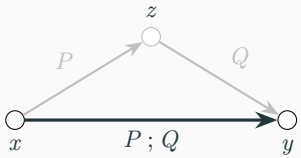



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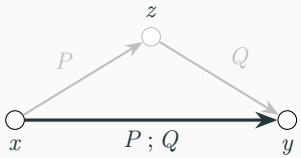

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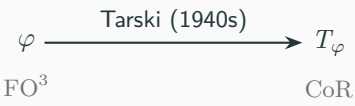



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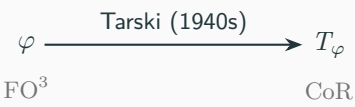

$|T_\varphi| = 2^{O(|\varphi|)}$ (classical)

$T_\varphi$: a smallest $\mathrm{CoR}$ term equivalent to $\varphi$ on finite structures

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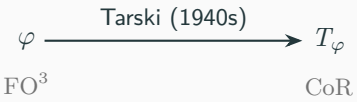

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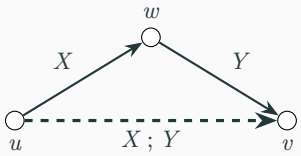
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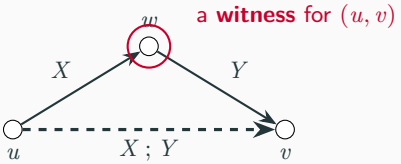
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$$X\,;\,Y \coloneqq \{(u,v) : \text{some } w \in \mathbb{D} \text{ satisfies } (u,w) \in X,\ (w,v) \in Y\}$$

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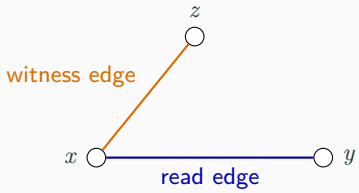



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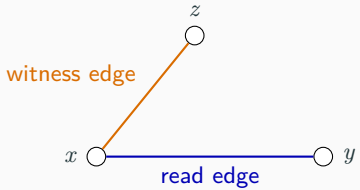
$$G_k(x, y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[ (P_i(x, y) \wedge Q_i(x, z)) \vee (Q_i(x, y) \wedge P_i(x, z)) \right]$$




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| | $i=1$ | $i=2$ | $i=3$ | $i=4$ |
|---|---|---|---|---|
| read edge | $P_1$ | $Q_2$ | $P_3$ | $Q_4$ |
| witness edge | $Q_1$ | $P_2$ | $Q_3$ | $P_4$ |

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# The DNF: $2^k$ compositions

$$\underbrace{H_\emptyset \cap (H_{[k]}\,;\mathbf{1}) \quad \cup \quad H_{\{1\}} \cap (H_{\{1\}^c}\,;\mathbf{1}) \quad \cup \quad H_{\{1,2\}} \cap (H_{\{1,2\}^c}\,;\mathbf{1}) \quad \cup \quad \cdots}$$

$$G_k \equiv \bigcup_{A\subseteq[k]} \left[H_A \cap (H_{A^c}\,;\mathbf{1})\right] \;:\; 2^k \textbf{ compositions}$$

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# Main results

**Thm. (Exponential lower bounds for full $\mathrm{CoR}$)**

*Every relational circuit $C \equiv_{\mathrm{fin}} G_k$ has $\geq 2^{k-1}$ composition gates.*

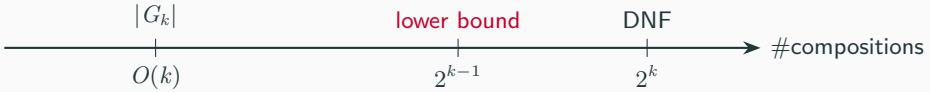



- shared subterms, complement, converse: all allowed and free
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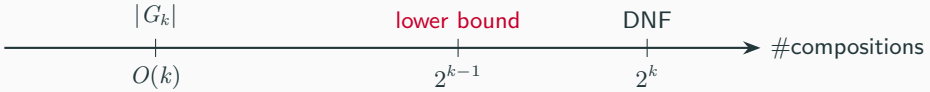



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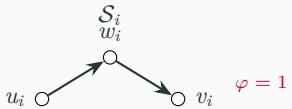
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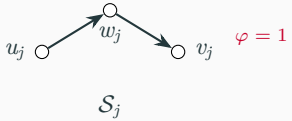



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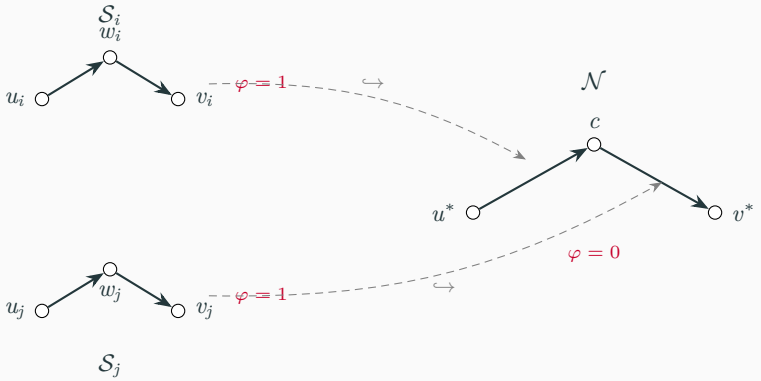
# $\Sigma_1\mathrm{CoR}$: a fooling-set argument (nakamura2020), (nakamura2022)



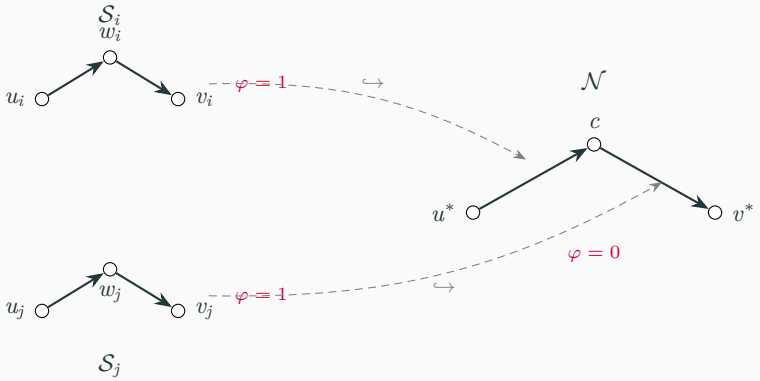

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1. complement reverses $(\diamond)$

2. $\overline{(E \cap (E\,;\overline{\mathbf{id}}))\,;\mathbf{1}}$ separates every $\mathcal{S}_i$ from $\mathcal{N}$ (two compositions)

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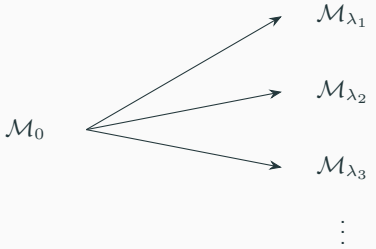
# Our technique: exact predictions on one reference structure

$\mathcal{M}_0$

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each composition notices at most **two** $\Rightarrow$ $2^k \leq 2 \cdot g_i(C)$

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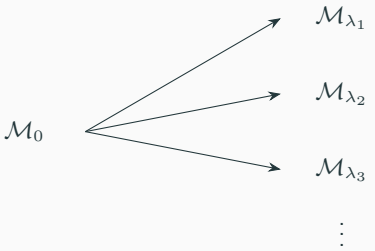



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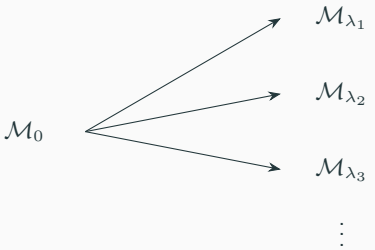



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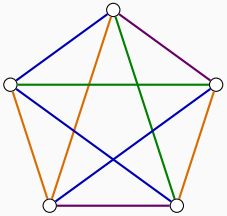
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# The reference structure



an edge-colored complete graph

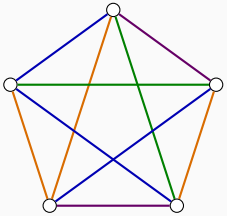
colors $\mu \subseteq [k]$; on an edge of color $\mu$:

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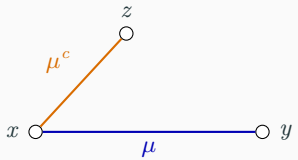
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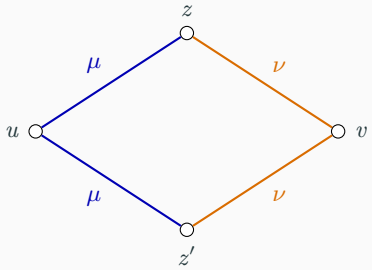
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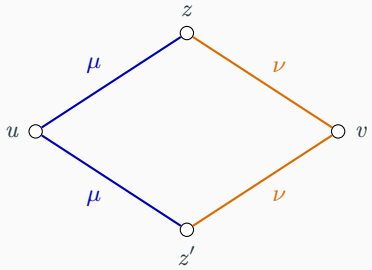



for all $u \neq v$ and all colors $\mu, \nu$: at least two witnesses

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fix one: the **reference structure** $\mathcal{M}_0$, and a vertex $\hbar$: the **hub**

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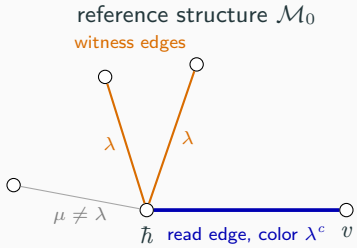



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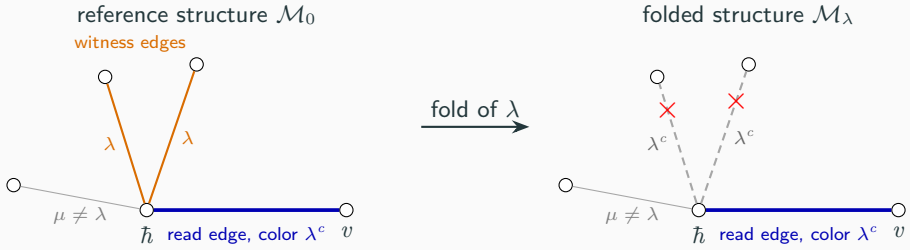

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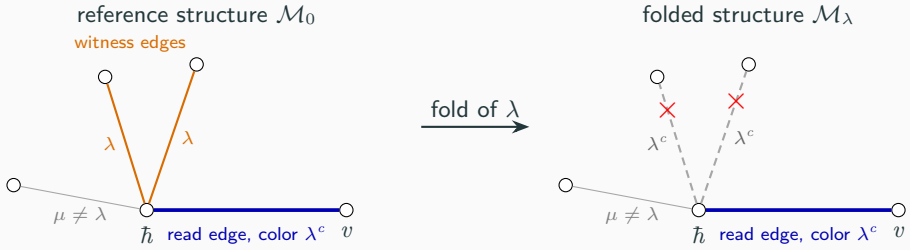

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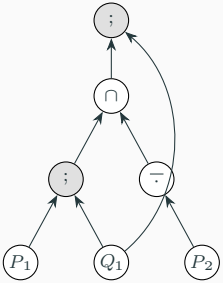

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a relational circuit

gate by gate: is $[\![g]\!]_{\mathcal{M}_\lambda}$ what $[\![g]\!]_{\mathcal{M}_0}$ **predicts**? $(\star)$

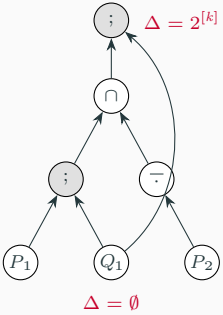
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Boolean operations and converse notice nothing

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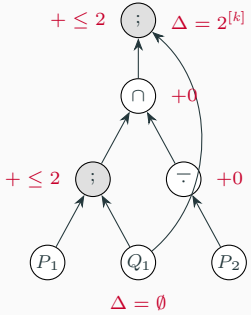
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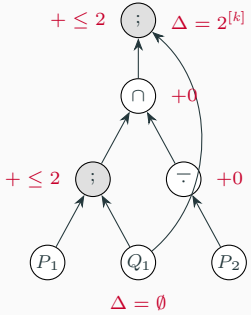
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$g$ **notices** $\lambda$: a new violation at $g$

Boolean operations and converse notice nothing

every fold must be noticed somewhere $\Rightarrow$ $2^k \leq 2 \cdot g_{;}(C)$ (also with sharing)

# Predictions and noticing




a relational circuit

gate by gate: is $[\![g]\!]_{\mathcal{M}_\lambda}$ what $[\![g]\!]_{\mathcal{M}_0}$ **predicts**? $(\star)$

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# Values on the reference structure are color tables

| $X$ | $X[\mathbf{id}]$ | $X[\{1\}]$ | $X[\emptyset]$ |
|---|---|---|---|
| $P_1$ | 0 | 1 | 0 |
| $Q_1$ | 0 | 0 | 1 |
| $P_1 \,;\, Q_1$ | 0 | 1 | 1 |
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$$X^{(\lambda)}(u,v) \coloneqq X[\chi_\lambda(u,v)] \qquad \text{(read the table of } X \text{ with the new colors)}$$

$$\overline{X^{(\lambda)}} = \overline{X}^{(\lambda)}, \quad X^{(\lambda)} \cup Y^{(\lambda)} = (X \cup Y)^{(\lambda)}, \quad \ldots$$

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**Lem. (Each composition notices at most two folds)**

*For color tables $X, Y$: $(X\,;\,Y)^{(\lambda)} = X^{(\lambda)}\,;\,Y^{(\lambda)}$ for all $\lambda$ except at most two.*



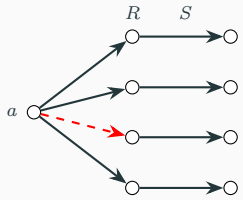

on an arbitrary structure: a single $R\,;\,S$ reacts to $t$ separate changes

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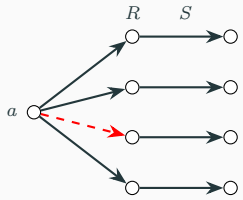

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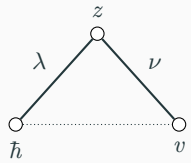



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# At the hub, and away from the hub

(a) $\mathcal{M}_0$, at $(\hbar, v)$



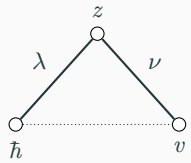

witness $z$: $(X\,;\,Y)(\hbar, v) = 1$

$X \coloneqq H_\lambda$, $Y \coloneqq H_\nu$ $(\nu \neq \lambda)$; noticed: the prediction $(X\,;\,Y)^{(\lambda)} = 1$ differs from $X^{(\lambda)}\,;\,Y^{(\lambda)} = 0$

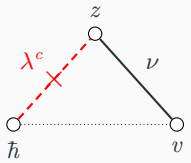
differ only if $S_X = \{\lambda\}$, $S_Y = \{\lambda\}$, or $S_X \cap S_Y = \{\lambda\}$ $\Rightarrow$ at most two $\lambda$

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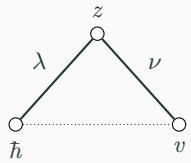

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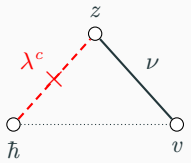
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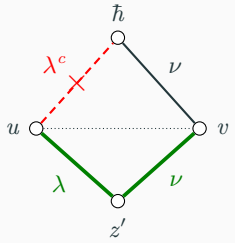

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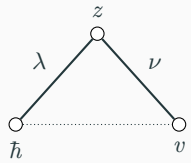

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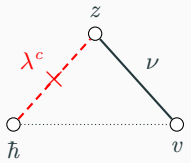
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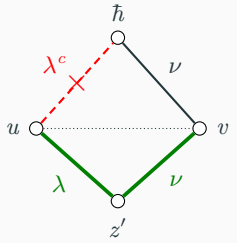

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# Outline

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| target or setting | folds noticed per gate |
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| relational circuits | composition 2 |
| $\mathrm{FO}^2$ formulas, sentences | quantifier 1 |
| matrix languages, breadth $\beta$ | aggregation $\leq 3\beta$ |
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